

[DC] Road-map to Efficient Attention Guided Augmented Reality User Interfaces: From Controlled Environments to the Wild

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ABSTRACT

This doctoral consortium paper discusses my roadmap towards efficient attention-guided Augmented Reality interfaces for public safety. My main goal is to evaluate the effect of AR interface placement and interaction design on performance, workload, and situational awareness under a dual-task configuration. Currently, I am working on how different AR display placements under dual-task configuration contribute to trade-offs between performance and workload. I aim to evaluate core AR UI modules, such as information display, path guidance, annotations, and alerts, using a dual-task evaluation toolkit, and work with first responders to validate the findings from the lab in a real-life scenario. The goal of this roadmap is to recommend practical design considerations and empirically validated AR interfaces to assist first responders in maintaining performance during dynamic scenarios.

Index Terms: Augmented Reality, First Responders, Dual-Task Performance, 3D User Interfaces, Public Safety

1 RESEARCH VISION

With the advent of Augmented Reality (AR) in mobile contexts, such as real-world way-finding or guidance, training, and assistance in various aspects, researchers have been investigating opportunities for first responders for AR guidance and assistance in decision-making, situational awareness, and information access tasks in dynamic situations. Existing research shows that, first responders exhibit a higher physical demand [1][2] than an average person, which combined with inherent dynamic decision-making cost to public safety tasks, raises concerns regarding attentional demands, as such a scenario leads to divided attention between both the physical world and the digital content provided by the AR display [3]. Hence, leading to increased cognitive load and a higher risk of performance degradation. Recent studies have identified the gaps [4] and potential opportunities [5] to address these concerns in terms of providing different Augmented Reality modules, such as adaptive user interfaces, cues for guidance, and spatial annotations, to assist in tasks performed by first responders.

As recent reviews indicate, the focus is on AR solutions rather than a rigorous evaluation of aspects such as interaction with AR interfaces while considering cognitive cost and stress, which are still relatively rare [4]. Recent studies have shown promising AR modules that have been studied to assist first responders in reducing their effort [5]. These include training scenarios such as traffic stop de-escalation [6], search and rescue, and an Emergency Medical Service (EMS) scenario [5]. However, none of the implementations have been rigorously evaluated, leaving a considerable gap in understanding the trend to balance cognitive load and awareness while maintaining performance and stability.

Recent studies have begun to reveal trends in the dual-task configurations, which combine the performance of physical and virtual

tasks simultaneously. This enables a more accurate analysis of the cost of attention and workload. Prior studies on adding a secondary task, such as memorization, to a common primary task, such as walking, have shown reduced performance and attention and increased gait instability [7]. However, presenting virtual content to the user in specific ways can either amplify or mitigate these effects [3].

In my Ph.D., I plan to evaluate core augmented reality user interface (UI) modules, including methods for displaying information and alerts, path guidance, and annotation tools, to provide design guidelines and attention-effective AR user interfaces for first responders while preserving task performance, perceived workload, and awareness. I aim to conduct a series of studies to evaluate AR UI modules in a lab-controlled environment using a dual-task configuration toolkit and use the combined findings to evaluate them in a more applied study with first responders in real-world scenarios.

2 ONGOING WORK

At the current stage of my PhD, I am focusing on the analysis of dual-task performance regarding AR display placement. A literature survey on reference frames and dual-task configurations has led me to this idea. Previous literature focuses mostly on tasks related to target selection [8], notification [9], and reading information [10] while performing a locomotion task, with the AR content being presented at different reference frames relative to the body (head, hand, and chest). This addresses the gap in task selection, where memory tasks are relatively less common, which may actually induce cognitive load [11]. However, a recent study [7] used a memory task and locomotion to consider dual-task performance and workload effects, but the task involved only visual feedback rather than interaction with the AR display.

The study that I am currently working on addresses the gaps in using an interactive memory task during locomotion with the common reference frames that have been used in the literature. Additionally, the AR display behavior varies relative to the body, so that the interface moves with the head, chest, or hand. This will allow me to find a contrast between different anchoring behaviors, both empirically and preference-wise. The memory task I chose is the Corsi Block Tapping test [12], modified to resemble a Simon Game, with locomotion involving walking with and without obstacles. The objective measures include task and gait performance, while the subjective measures comprise NASA-TLX, custom preference ranking, and the Simulator Sickness Questionnaire to better understand the trade-off between performance and workload under a dual-task configuration.

The study has received IRB approval, and a pilot study has been completed; full data collection will commence from January 2026..

3 FUTURE RESEARCH DIRECTIONS

Following my current study, I look forward to progressing my PhD journey along two tightly connected directions:

3.1 Identifying and evaluating most commonly used AR user interfaces related to public safety:

The first steps towards this direction would be to understand which AR interfaces are most relevant to public safety and how specific

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design considerations affect performance, workload, and situational awareness. I plan to focus on a small set of recurring interfaces that have been proposed in the literature, including AR display placement, navigation guidance, annotation tools, and relaying notifications. I aim to utilize this design space to evaluate and investigate how various AR interfaces impact performance, workload, and situational awareness when people move and perform a task simultaneously, considering different placements, visualizations, and behaviors.

Regarding my plan, I aim to use a consistent evaluation toolkit, which includes a dual-task configuration that combines locomotion with a cognitively demanding task. This configuration will utilize objective measures of task performance and gait metrics, as well as subjective measures related to workload, preference, and situational awareness. This will allow coherence between studies regarding the design trade-offs related to performance and workload. The outcome of this plan will yield empirically grounded design recommendations for attention-efficient AR interfaces in public safety.

3.2 Connecting and validating the findings with first responders in realistic scenarios to demonstrate real-world impact:

The second phase of my plan involves validating the most promising AR interfaces from the lab in a more applied scenario with first responders in operational settings. Controlled studies are ideal for isolating variables, but they do not fully capture real-world scenarios. Therefore, taking the concept of 'from lab to the wild' [13], I plan to combine the individual aspects from phase 1 into a scenario-based training exercise and evaluate them based on performance, usability, and trust.

There are two important reasons for this phase. Firstly, interfaces evaluated and designed under lab conditions can actually help first responders in situations where AR technology is meant to address, rather than focusing on abstract tasks. Second, it will act as a feedback loop; taking feedback from first responders, both qualitatively and quantitatively, can lead to new opportunities for design and evaluation methods that can be considered in a controlled environment.

Together with these two plans, I aim to lead a systematic evaluation with end users, which will uphold both scientific contributions related to attention and workload in AR and provide practical guidance for AR interfaces that will benefit public safety personnel.

4 CONCLUSION

The expected contributions from my dissertation are empirical evidence on how different AR modules, such as display placement, navigation guidance, and methods of notification relays, affect the dual-task paradigm in terms of performance, workload, and situational awareness in the public-safety context. Also, providing design considerations and guidelines for efficient AR interfaces, taking them from a controlled environment to the wild, thus validating empirical findings from the lab.

Regarding the IEEE VR 2026 Doctoral Consortium, I seek feedback on the scope and logical validation of the research agenda as a PhD dissertation topic and the correctness of the dual-task configurations, as well as the method for empirical validation to ensure meticulous findings for the final integrated application for the first responders. Lastly, guidance on study scenarios, measures, or analysis methods will be very helpful to overcome any shortcomings for the study to be rigorous.

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